# A positive local representation and an unresolved pentagon reproduction

Alan Day Chengheng*

Working research note — 1 November 2026

**Abstract**

We follow the existing positive-cosmological-constant quantization of Kim–Scarinci to identify a concrete local Hilbert-space benchmark for the missing observer gluing problem. We give an explicit four-puncture Heisenberg representation, distinguish phase pairing from a modulus square, and implement the stated pentagon in two discretizations. Scalar controls agree, but our non-flat operator reproduction remains unresolved. This is a diagnostic record, not a refutation of the continuum theorem or a construction of a finite-$G$ observer dual.

## 1 What is known, and what is being tested

CLP's fixed-topology continuation does not itself supply a general de Sitter gluing theory; its discussion points to both quantization and the non-diagonal physical Gram problem [1, 2]. The KS construction supplies explicit unitary local operators on punctured-surface Hilbert spaces [3, 4]. Its full flip also contains a linear, or Shale–Weil, intertwiner. Here we test the separately stated two-variable pentagon, not a complete flip network.

Use dimensionless coordinates, $\Lambda = +1$, and $\hbar > 0$. The proposed comparison $b^2 = i/\kappa = i\hbar$ matches deformation parameters; it is not an observer mass dictionary. The source functions are

$$\psi_q(w) = \prod_{n=0}^{\infty} (1 + q^{2n+1} w)^{-1}, \tag{1}$$

$$\Phi^{i\hbar}(z) = \frac{\psi_{e^{-\pi\hbar}}(e^z)}{\psi_{e^{-\pi/\hbar}}(e^{z/(i\hbar)})}, \qquad z = x + i\hbar y, \tag{2}$$

$$F_\hbar(x, y) = \Phi^{i\hbar}(z)\Phi^{-i\hbar}(\bar{z}) = \frac{\Phi^{i\hbar}(z)}{\overline{\Phi^{i\hbar}(z)}}. \tag{3}$$

Consequently $|F_\hbar| = 1$ almost everywhere. Isolated exceptional points can be assigned arbitrary unit values for the Lebesgue multiplication operator. This phase ratio is distinct from $|\Phi|^2$. Neither a scalar modulus square nor scalar positivity identifies an inner product for general mixed four-point pairings.

*AI research assistant (OpenAI).

## 2 An explicit four-puncture representation

Order the edges as $12, 13, 14, 23, 24, 34$, and orient faces as $(132), (124), (143), (234)$. Let $V_{ep} = 1$ when $p$ is an endpoint of $e$, and zero otherwise. The oriented exchange matrix is

$$E = UJU^T, \qquad U = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -1 & -1 \\ -1 & -1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \tag{4}$$

Here $\operatorname{rank} E = 2$, $\operatorname{rank} V = 4$, $EV = 0$ and $V^T U = 0$. On $\mathcal{H} = L^2(\mathbb{R}^2, dt_1 dt_2)$ put

$$p_a = -\pi i \partial_{t_a}, \qquad \mathbf{x} = U\mathbf{p}, \qquad \mathbf{y} = UJ\mathbf{t}. \tag{5}$$

On Schwartz functions, and in the Weyl sense after closure,

$$[x_i, x_j] = [y_i, y_j] = 0, \quad [x_i, y_j] = \pi i E_{ij}, \quad V^T\mathbf{x} = V^T\mathbf{y} = 0. \tag{6}$$

For example $(x_1, y_1) = (p_1, t_2)$ and $(x_2, y_2) = (p_2, -t_1)$. Each same-edge pair strongly commutes. The four operators on these two edges recover two canonical pairs, so this representation is irreducible. This is an explicit instance of the known constraint reduction, not a new quantization theorem.

There is also a useful purely algebraic extension. Define

$$R = V(V^T V)^{-1}, \qquad \mathbf{x}_\xi = U\mathbf{p} + R\xi, \qquad \mathbf{y}_\eta = UJ\mathbf{t} + R\eta, \qquad \xi, \eta \in \mathbb{R}^4. \tag{7}$$

Then $V^T\mathbf{x}_\xi = \xi$ and $V^T\mathbf{y}_\eta = \eta$, with the same commutators, self-adjointness and local unitarity. This does *not* establish that these central characters are conical masses, nor construct all intertwiners between triangulations.

**Relation to the CLP special function.** With $b^2 = i\hbar$, $Q = b + b^{-1}$ and $w = 2\pi i b\zeta$, CLP (3.48) becomes

$$S_b(Q/2 + \zeta) = \Phi^{i\hbar}(w) \exp\left[-\frac{\pi i}{2}\zeta^2 + \frac{i\pi}{24}(Q^2 - 2)\right]. \tag{8}$$

Thus the same compact products occur, but the Gaussian prefactor and pairing operation must be retained. Equality of special-function ingredients does not identify their states or gluing measures. CEM's $h + \tilde{h}^* = 1$ applies to weights in one wavefunction; it should not be confused with CLP's paired-factor $h_+ + h_- = 1$.

## 3 A mixed Gram test and the pentagon

For calculation it is convenient to use the unitarily equivalent realization

$$(x_1, y_1) = (x, y), \qquad (x_2, y_2) = (p_y, p_x), \qquad p_x = -\pi i \partial_x, \quad p_y = -\pi i \partial_y. \tag{9}$$

It has $[x_1, y_2] = [y_1, x_2] = \pi i$ and all other required commutators zero. Let

$$A = F_\hbar(x, y), \quad B = F_\hbar(p_y, p_x), \quad M = e^{ixy/\pi}, \quad C = M^{-1} B M. \tag{10}$$

The exact differential identities $M^{-1} p_y M = p_y + x$ and $M^{-1} p_x M = p_x + y$ give

$$C = F_\hbar(x + p_y, y + p_x). \tag{11}$$

The source pentagon (KS v2, Proposition 6.15; Kim v2, (3.8)) is

$$AB = BCA = BM^{-1}BMA. \tag{12}$$

Our numerical residual is the norm of the difference of these two words acting on the normalized state

$$\psi(x, y) = \pi^{-1} \exp[-(x^2 + y^2)/(2\pi)]. \tag{13}$$

No observer time or physical OTOC is assigned to these operators.

The different words $AB$ and $BA$ are used for a separate Gram check. For $u = AB\psi$, $v = BA\psi$ and $z = \langle u, v \rangle$,

$$G = \begin{pmatrix} 1 & z \\ \bar{z} & 1 \end{pmatrix} \succeq 0, \qquad D = \|u - v\|^2 = 2 - 2 \operatorname{Re} z. \tag{14}$$

This follows from the stated Hilbert space and unitarity; it does not imply (12). In particular, a discretization made of unitary matrices can pass every norm/Gram check while failing its continuum operator target.

**Two implementations.** The periodic FFT uses nodes $x_j = -L + 2Lj/N$ and a two-dimensional unitary Fourier transform. Its momentum grid is

$$p_k = 2\pi^2 \operatorname{fftfreq}(N, 2L/N).$$

The Hermite DVR uses eigenvectors $Q_{nj}$ of the Jacobi matrix with off-diagonal entries $\sqrt{n/2}$. Its physical nodes are $\sqrt{\pi}$ times the Hermite nodes and its Fourier matrix is

$$T = Q^T \operatorname{diag}((-i)^n) Q. \tag{15}$$

Both implement $B = T^\dagger F_\hbar(p_y, p_x) T$ and the same chirp factorization, with the same continuum Gaussian target. Finite matrices do not satisfy exact canonical Weyl relations.

## 4 Results: scalar agreement, operator residual

There are 21 matched-protocol runs: three FFT grids, four Hermite orders, and three multipliers at each setting. The flat control uses $F_0(x,y) = \exp[-iy\log(1+e^x)/\pi]$; it is the independently defined $\Lambda = 0$ formula, not a claimed numerical $\hbar \to 0$ limit.

| Method | $\hbar = .5$ | $\hbar = 1$ | flat control |
|---|---|---|---|
| FFT $N = 256, L = 16$ | .03005165 | .12331190 | $1.40\,10^{-5}$ |
| FFT $N = 512, L = 24$ | .02975935 | .12189770 | $8.54\,10^{-9}$ |
| FFT $N = 1024, L = 32$ | .02955658 | .12168478 | $3.01\,10^{-12}$ |
| Hermite $N = 128$ | .02940377 | .12229173 | $1.52\,10^{-10}$ |
| Hermite $N = 256$ | .02948460 | .12170210 | $4.77\,10^{-15}$ |
| Hermite $N = 512$ | .02946719 | .12156956 | $4.81\,10^{-15}$ |

Entries are $\|(AB - BM^{-1}BMA)\psi\|$. The non-flat reproduction is unresolved; we do not mark it as passed. At Hermite $N = 512, \hbar = 1$ the order-exchange overlap is approximately $.94417241 - .00563230i$, giving $D = .11165518$. These are results of the discrete benchmark, not finite-$G$ observer correlations.

Thirty-six scalar/algebra checks passed. They include independent 55-digit compact products, the inversion identity

$$F_\hbar(x,y)F_\hbar(-x,-y) = e^{-ixy/\pi}, \tag{16}$$

three slanted Barnes-integral controls, exact puncture matrices, and the chirp conjugation. The Barnes line is $p = e^{-i\pi/4}(t + 0.2i)$ with $-30 < t < 30$ at $\hbar = 1$. This quadrature has empirical, not interval-certified, errors.

For a compact-product logarithm, truncate after $n = N - 1$ and set $r_0 = e^{a-(2N+1)r} < 1$. Its exact-arithmetic logarithmic tail obeys

$$\left|\sum_{n=N}^{\infty} \log(1 + e^{a-(2n+1)r+ib})\right| \le \frac{r_0}{(1-e^{-2r})(1-r_0)}. \tag{17}$$

Twice the sum of the two tails bounds the scalar phase error. It does not bound floating-point errors, aliasing, or Hermite functional-calculus errors.

The 161-check evidence replay verifies hashes, run counts and Gram arithmetic while explicitly retaining the unresolved status. Earlier prototypes used different states/representations and are preserved separately. Neither their norm preservation nor the agreement between two discretizations is a continuum counterexample.

## 5 A controlled next test

The next calculation should replace nodewise DVR functional calculus by actual Hermite Galerkin matrix elements. This matters because a DVR phase multiplier is exactly unitary at every finite order, concealing the leakage of a projected continuum state.

Here is an elementary error ledger for that task. Let $P$ be a finite orthogonal Hermite projector, $U_j$ the actual bounded unitary multiplication or Fourier-conjugated multiplication operators, and $T_j = PU_jP$. For $\psi \in P\mathcal{H}$, define $v_0 = \psi$, $v_j = T_j v_{j-1}$. Then

$$\ell_j^2 = \|(1-P)U_j v_{j-1}\|^2 = \|v_{j-1}\|^2 - \|v_j\|^2, \tag{18}$$

$$\|U_m \cdots U_1 \psi - v_m\| \le \sum_{j=1}^{m} \ell_j. \tag{19}$$

The proof is a one-step telescoping triangle inequality and orthogonality of $P$. It requires true compressed operators; substituting a unitary DVR matrix makes the leakage estimate invalid.

If computed matrices $\widetilde{T}_j$ satisfy $\|\widetilde{T}_j - T_j\| \le \epsilon_j$, the difference of their products is bounded by

$$\|\widetilde{T}_m \cdots \widetilde{T}_1 - T_m \cdots T_1\| \le \sum_{j=1}^{m} \epsilon_j \prod_{k<j} (1+\epsilon_k), \tag{20}$$

since $\|T_j\| \le 1$. A finite observed residual $r$ can lower-bound the continuum residual only after both word errors are controlled:

$$\|(AB - BM^{-1}BMA)\psi\| \ge r - \delta_{\text{left}} - \delta_{\text{right}}. \tag{21}$$

This is an actionable certification target, not a completed estimate. Scalar zero/pole neighborhoods must be excised with norm bounds or integrated with appropriate local coordinates. The Gaussian tails, quadrature errors and Hermite leakage must all be recorded.

**Open physical interfaces.** Even a successful continuum reproduction would leave the mass and holonomy map, boundary states, full triangulation intertwiners, and the physical observer insertions and Gram gluing unresolved. Conversely, the current numerical difficulty is not evidence that de Sitter holography is impossible. We keep the original scientific objective open.

## References


[1] J. Chandra, D. Liska and B. Post, *3D de Sitter from complex Virasoro TQFT: Observers and knots*, arXiv:2610.30247v1.

[2] S. Collier, L. Eberhardt and B. Mühlmann, *A microscopic realization of $dS_3$*, arXiv:2501.01486v1.

[3] H. K. Kim and C. Scarinci, *A quantization of moduli spaces of 3-dimensional gravity*, arXiv:2112.13329v2.

[4] H. K. Kim, *A trilogy of mapping class group representations from three-dimensional quantum gravity*, arXiv:2308.02934v2.